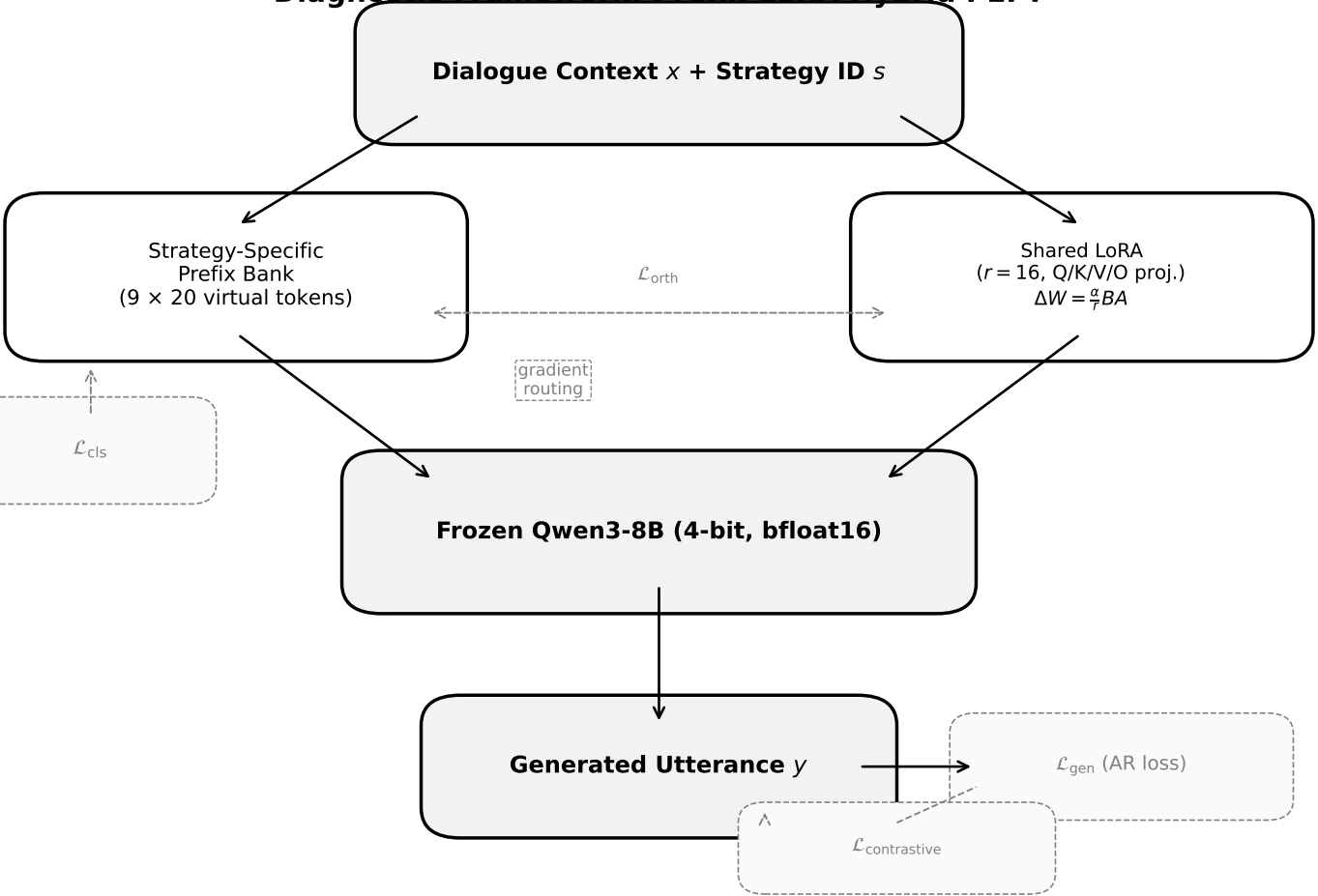


Diagnostic Framework: Prefix-LoRA Hybrid PEFT



$\mathcal{L}_{orth}$ : Orthogonal constraint (B5, B6 variants)  
 $\mathcal{L}_{cls}$ : Auxiliary strategy classification (B6)  
**Gradient routing:** gen → LoRA, cls → Prefix (B6<sub>grad</sub>)  
 $\mathcal{L}_{contrastive}$ : Token-level strategy sensitivity (B6<sub>contrast</sub>)